



INNATE™ ST100

Precision Packaging Resin

Overview

INNATE™ ST100 Precision Packaging Resin is designed for exceptional abuse performance at a medium density. It offers a unique combination of high stiffness with excellent toughness which enables new performance levels for all polyethylene flexible packaging. This product also delivers potential for downgauging, and it is easy to process at typical blown film lines.

Complies with:

- U.S. FDA 21 CFR 177.1520(c) 3.2a
- European Commission Regulation (EU) No 10/2011

Consult the regulations for complete details

Additive

- Antiblock: No
- Slip: No
- Processing Aid: No

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.928 g/cm³	0.928 g/cm³	ASTM D792
Melt Index (190°C/2.16 kg)	0.85 g/10 min	0.85 g/10 min	ASTM D1238
Films	Nominal Value (English)	Nominal Value (SI)	Test Method
Film Thickness - Tested	2.0 mil	51 µm	
Film Puncture Energy	45.5 in·lb	5.14 J	
Film Puncture Force	18.6 lbf	82.7 N	
Film Puncture Resistance	151 ft·lb/in³	12.5 J/cm³	Dow Method
Secant Modulus			ASTM D882
2% Secant, MD	50300 psi	346 MPa	
2% Secant, TD	56300 psi	388 MPa	
Tensile Strength			ASTM D882
MD : Yield	2170 psi	15.0 MPa	
TD : Yield	2180 psi	15.0 MPa	
MD : Break	4660 psi	32.1 MPa	
TD : Break	4940 psi	34.1 MPa	
Tensile Elongation			ASTM D882
MD : Yield	470 %	470 %	
TD : Yield	540 %	540 %	
Dart Drop Impact ¹	2100 g	2100 g	ASTM D1709
Elmendorf Tear Strength			ASTM D1922
MD	320 g	320 g	
TD	790 g	790 g	
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Vicat Softening Temperature	235 °F	113 °C	ASTM D1525
Melting Temperature	256 °F	124 °C	DSC
Optical	Nominal Value (English)	Nominal Value (SI)	Test Method
Haze	61.0 %	61.0 %	ASTM D1003

Extrusion Notes

Fabrication Conditions for Blown Film:

- Die Diameter: 8 in.
- Screw Type: DSB II
- Die Gap: 70 mil (1.7 mm)
- Melt Temperature: 433 °F
- Output: 6.2 lb/hr/in of die circumference
- Screw Size: 3.5 in.
- Blow-up Ratio: 2.5 to 1
- Screw Speed: 42 rpm
- Frost Line Height: 37 in

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Method A

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Additional Information

North America

U.S. & Canada:

1-800-441-4369

1-989-832-1426

Mexico:

+1-800-441-4369

Latin America

Argentina:

+54-11-4319-0100

Brazil:

+55-11-5188-9000

Colombia:

+57-1-219-6000

Mexico:

+52-55-5201-4700

Europe/Middle East

+800-3694-6367

+31-11567-2626

Italy:

+800-783-825

South Africa

+800-99-5078

Asia Pacific

+800-7776-7776

+603-7965-5392

www.dow.com

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